

1. Definitions

Let's start with our polygons which have vertices $\vec{V}_k$ for $k \in [0, N_v)$ where N_v is the total number of vertices across all polygons in the packing. Let there be N polygons so that polygon k has N_k vertices. Then $N_v = \sum_{k=1}^N N_k$. Let's also let

$$\vec{e}_k \equiv \vec{V}_{z(k)} - \vec{V}_k \quad \text{eq 1.1}$$

and

$$\vec{b}_k \equiv \vec{V}_{z(k)} - \vec{V}_{z'(k)} \quad \text{eq 1.2}$$

also

$$A = \frac{1}{2} \sum_{k=0}^{n-1} V_{z(k)}^\alpha \epsilon_{\alpha\beta} V_k^\beta \quad \text{eq 1.3} \quad \text{Area of a polygon.}$$

The perimeter is

$$p = \sum_{k=0}^{n-1} |\vec{e}_k| \quad \text{eq 1.4}$$

Let's also be very clear that the vertices are ordered in a ccw fashion

2 Intersections

Obviously

$\vec{V}_k + \vec{e}_k = \vec{V}_{z(k)}$ by definition so

$$\boxed{\vec{P}_k(s) \equiv \vec{V}_k + s \vec{e}_k \text{ for } s \in [0, 1]} \quad \text{eq 2.1}$$

interpolates linearly between the two points.

For two edges which intersect, i and j ,
we have that $\exists s, t \in [0, 1]$ st

$\vec{P}_i(s) = \vec{P}_j(t)$. Rearranging gives

$$V_i^x - V_j^x + s e_i^x - t e_j^x = 0$$

$$\frac{V_i^x - V_j^x + s e_i^x}{e_j^x} - \frac{V_i^y - V_j^y + s e_i^y}{e_j^y} = 0$$

$$\frac{V_i^x - V_j^x + S e_i^x}{e_j^x} - \frac{V_i^y - V_j^y + S e_i^y}{e_j^y} = 0$$

$$S \left(\frac{e_i^x}{e_j^x} - \frac{e_i^y}{e_j^y} \right) = - \frac{V_i^x - V_j^x}{e_j^x} + \frac{V_i^y - V_j^y}{e_j^y}$$

$$S(e_i^x e_j^y - e_j^x e_i^y) = -[(V_i^x - V_j^x) e_j^y - (V_i^y - V_j^y) e_j^x]$$

$$S e_i^\alpha \epsilon_{\alpha\beta} e_j^\beta = (\vec{V}_j - \vec{V}_i)^\alpha \epsilon_{\alpha\beta} e_j^\beta$$

$$S = \frac{(\vec{V}_j - \vec{V}_i)^\alpha \epsilon_{\alpha\beta} e_j^\beta}{e_i^\alpha \epsilon_{\alpha\beta} e_j^\beta} \quad \text{eq 2.2}$$

and

$$t = \frac{(\vec{V}_i - \vec{V}_j)^\alpha \epsilon_{\alpha\beta} e_i^\beta}{e_j^\alpha \epsilon_{\alpha\beta} e_i^\beta} \quad \text{eq 2.3}$$

by symmetry.

3 Area of Overlap

Let there be two intersecting polygons A and B.

The area of intersection between A and B is defined as

$$|A \cap B| \equiv \frac{1}{2} \epsilon_{\alpha\beta} \oint_{\partial(A \cap B)} X^\alpha dX^\beta \quad \text{eq 3.1}$$

for the boundary of the polygon formed by the overlaps given by $\partial(A \cap B)$.

As you traverse the shape formed by this overlap, the vertices come in 4 flavors =

1. $\vec{V}_k \notin \partial(A \cap B)$, $\vec{V}_{z'(k)} \notin \partial(A \cap B)$
2. $\vec{V}_k \notin \partial(A \cap B)$, $\vec{V}_{z'(k)} \in \partial(A \cap B)$
3. $\vec{V}_k \in \partial(A \cap B)$, $\vec{V}_{z'(k)} \notin \partial(A \cap B)$
4. $\vec{V}_k \in \partial(A \cap B)$, $\vec{V}_{z'(k)} \in \partial(A \cap B)$

We may have the case where the number of vertices on the boundary $\partial(A \cap B)$ is very large. There's a way to combat this, but first, we consider an intersection between edges i and j to be denoted $\{i, j\}$ where $\exists s, t$ such that

point $\vec{v}_i + s \vec{e}_i \in \partial(A \cap B)$ but $\vec{v}_i + t \vec{e}_i \notin \partial(A \cap B)$ for $s > t$.

and similarly for j except $s < t$.

If you follow the boundary starting at the intersection and following the boundary of the polygon associated with edge i , then eventually you will reach another intersection $\{k, l\}$ where k follows j and l follows i . Denote this intersection pair as $\{i, j; k, l\}$.

For an edge from $\vec{P}$ to $\vec{Q}$, we define the area contribution as

$$h(\vec{P}, \vec{Q}) \equiv \frac{1}{2} (\vec{P} - \vec{R})^\alpha \epsilon_{\alpha\beta} (\vec{Q} - \vec{R})^\beta$$

eq 3.2

where

$\vec{R}$ is a reference point. We usually use the lowest indexed vertex in the lower indexed polygon between A and B.

Note that there are exactly as many pairs of intersections as intersections. Every intersection has a follower.

Our total overlap area is then

$$U = \sum_{\{i,j;k,l\}} \left\{ \delta_{il} h(\vec{f}_{ij}, \vec{f}_{kl}) + (1 - \delta_{il}) \left[h(\vec{f}_{ij}, \vec{v}_{z(i)}) + h(\vec{v}_l, \vec{f}_{kl}) + \sum_{m=z(i)}^{z'(l)} h(\vec{v}_m, \vec{v}_{z(m)}) \right] \right\}, \quad \text{eq 3.3}$$

We can clean this up a little bit by defining a coordinate with a fractional or continuous index. Let i have a fractional part corresponding to the portion of the edge which has been traversed. Then we have

$$\vec{q}_i = \vec{v}_{i^-} + (i - i^-) \vec{e}_{i^-} \quad \text{and}$$

$$U = \sum_{\{i,j;k,l\}} \left\{ \delta_{i \neq l} h(\vec{q}_i, \vec{q}_l) + (1 - \delta_{i \neq l}) \left[h(\vec{q}_i, \vec{v}_{z(i^-)}) + h(\vec{v}_{z(l^-)}, \vec{q}_l) + \sum_{m=z(i^-)}^{z(l^-)} h(\vec{v}_m, \vec{v}_{z(m)}) \right] \right\}, \quad \text{eq 3.4}$$

Let the intersections be

$\{S_{i^-}, S_{j^-}, j, i\}$ in sorted form from

left to right for shape S_{i^-} and S_{j^-} .

We then search for $\{S_{j^-}, S_{i^-}, l, k\}$ via binary search to find the corresponding "next" intersection. We can just save that intersection's index instead of keeping another huge list.

We can then define $\delta_{mp} = 1$ when an edge m is fully between intersections $\{i, j\}$ and $\{k, l\}$ for $p = \{i, j; k, l\}$. If $\delta_{mp} = 0$ otherwise.

We can determine all p where $\delta_{mp} = 1$ via binary search (or linear search).

We then have

$$U_{\text{ex}} = \sum_{\{i, j; k, l\}} \left\{ \delta_{ix} h(\vec{a}_i, \vec{a}_l) \right.$$

$$\left. + (1 - \delta_{ix}) [h(\vec{a}_i, \vec{v}_{z(i)}) + h(\vec{v}_l, \vec{a}_l)] \right\}$$

$$U_{\text{int}} = \sum_{m=0}^{N_v-1} \sum_{p=\{i, j; k, l\}} \delta_{pm} h(\vec{v}_m, \vec{v}_{z(m)})$$

$$U = U_{\text{int}} + U_{\text{ex}} \quad .$$

eq 3.5

4 Forces on the Boundary

Differentiating the integral over the boundary has a trick associated with it related to the Hadamard structure theorem.

We can get out of computing the derivative of the intersection point (thank God).

We can take

$A_1 = \frac{1}{2} \oint_{\partial\Omega} X^\alpha E_{\alpha\beta} dX^\beta$ for the boundary Ω and consider perturbing

$\vec{X} \mapsto \vec{X} + \delta \vec{X}$ so that

$$\delta A_1 = \frac{1}{2} \oint_{\partial\Omega} \delta X^\alpha E_{\alpha\beta} dX^\beta + \frac{1}{2} \oint_{\partial\Omega} X^\alpha E_{\alpha\beta} d(\delta X^\beta)$$

Using integration by parts on the second term gives

$$\delta A_1 = \frac{1}{2} \oint_{\partial \Omega} \delta X^\alpha E_{\alpha\beta} dX^\beta + \frac{1}{2} X^\alpha E_{\alpha\beta} \delta X^\beta \Big|_{\Omega^-}^{\Omega^+} \\ - \frac{1}{2} \oint_{\partial \Omega} \delta X^\beta E_{\alpha\beta} dX^\alpha$$

and since $E_{\alpha\beta} = -E_{\beta\alpha}$, we have

$$\boxed{\delta A_1 = \oint_{\partial \Omega} \delta X^\alpha E_{\alpha\beta} dX^\beta.} \quad \text{eq 4.1}$$

We then consider let

$$\vec{X}_m(\lambda) = \vec{V}_m + \lambda \vec{E}_m \quad \text{for a point on}$$

edge m so that

$$\delta \vec{X}_m(\lambda) = (1-\lambda) \delta \vec{V}_m + \lambda \delta \vec{V}_{2(m)}$$

and

$$d\vec{X}_m = \vec{E}_m d\lambda$$

Then we have

$$\begin{aligned}\delta A_1 &= \sum_m \sum_{\alpha} \int_{\lambda_0^m}^{\lambda_f^m} \left[(1-\lambda) \delta V_m^\alpha + \lambda \delta V_{z(m)}^\alpha \right] \epsilon_{\alpha\beta} e_m^\beta d\lambda \\ &= \sum_m \sum_{\alpha} \left[\left(\lambda - \frac{1}{2} \lambda^2 \right) \right]_{\lambda_0^m}^{\lambda_f^m} \delta V_m^\alpha + \frac{1}{2} \lambda^2 \Big|_{\lambda_0^m}^{\lambda_f^m} \delta V_{z(m)}^\alpha \\ &\quad \times \epsilon_{\alpha\beta} e_m^\beta\end{aligned}$$

$$\begin{aligned}&= \sum_m \sum_{\alpha} \left[(\lambda_f - \lambda_0) - \frac{1}{2} (\lambda_f^2 - \lambda_0^2) \right] \delta V_m^\alpha \\ &\quad + \left[\frac{1}{2} (\lambda_f^2 - \lambda_0^2) \delta V_{z(m)}^\alpha \right] \epsilon_{\alpha\beta} e_m^\beta\end{aligned}$$

Let $\bar{S}_m = \frac{1}{2}(\lambda_f + \lambda_0)$ and $\Delta S_m = \lambda_f - \lambda_0$

so

$$\delta A_1 = \sum_m \sum_{\alpha} \left[\Delta S_m (1 - \bar{S}) \delta V_m^\alpha + \Delta S_m \bar{S} \delta V_{z(m)}^\alpha \right] \epsilon_{\alpha\beta} e_m^\beta$$

$$\delta A_1 = \sum_m \sum_{\alpha} \left[(1 - \bar{S}) \delta V_m^\alpha + \bar{S} \delta V_{z(m)}^\alpha \right] \epsilon_{\alpha\beta} \Delta S_m e_m^\beta$$

Eq 4.2

Now we can write this in terms of Kronecker δ as

$$\delta A_1 = \sum_m \sum_\alpha \sum_k \left[(1 - \bar{S}_m) \delta_{mk} + \bar{S}_m \delta_{z(m)k} \right] \delta V_k^\alpha \times E_{\alpha\beta} \Delta S_m e_m^\beta$$

Thus we have

$$\delta A_1 = \sum_k \sum_\alpha \left[\sum_m \left((1 - \bar{S}_m) \delta_{mk} + \bar{S}_m \delta_{z(m)k} \right) \Delta S_m e_{\alpha\beta} e_m^\beta \right] \times \delta V_k^\alpha$$

and since

$$\delta A_1 = \sum_k \sum_\alpha \frac{\partial A_1}{\partial V_k^\alpha} \delta V_k^\alpha,$$

We can conclude that

$$\frac{\partial A_1}{\partial V_k^\alpha} = \sum_m \left[(1 - \bar{S}_m) \delta_{mk} + \bar{S}_m \delta_{z(m)k} \right] \Delta S_m E_{\alpha\beta} e_m^\beta$$

eq 4.3

5 Intersections

Consider an intersection

$\{i, j\}$ which intersect at point

$$\vec{V}_i + S_i \vec{E}_i = \vec{V}_j + S_j \vec{E}_j.$$

Let's also consider the polygon index
 $p_i, p_j \in [0, N)$.

WLOG, $\vec{V}_i + (S_i + \epsilon) \vec{E}_i$ is on the overlap
boundary for some $\epsilon > 0$.

Sort all of the intersections by

$$\{p_i, p_j, i, S_j, i, S_i\}.$$

Then we find the intersection pair.

For intersection

$I_\alpha = \{p_i, p_j, i, s_i, j, s_j\}$, we want

to find intersection

$I_\beta = \{p_k, p_l, l, s_l, k, s_k\}$

We want to find

$p_k = p_j$ and $p_i = p_l$ which

should be a short list and we can find its boundaries with binary search (in general it has 2 entries). We then choose β so that $|i + s_i - (l + s_l)|$ is minimal over all polygons.

Next we loop over all vertices and check all intersections of that shape to see if vertex V is between $i+S_i$ and $l+S_e$. where $V \neq l$. If it is, you can compute the components necessary for its gradient and energy.

6. Mollification

Unfortunately, this alone doesn't work.

FIRE chokes and may or may not reach the basin. It certainly doesn't reach the minimum. We can't use a higher order method because the forces aren't continuous meaning the Hessian is ill-defined.

We can resolve both issues by mollifying the boundaries.

Consider the area of overlap,

$$A_{\cap} = \int_{\mathbb{R}^2} \mathbb{1}_A \mathbb{1}_B d^2x \quad 6.1$$

We can consider a kernel K_σ which smoothes out the edges so that instead of a step boundary, we have a smooth boundary. Applying this to one of the polygons gives

$$A_n^\sigma = \int_{\mathbb{R}^2} \mathbb{1}_A(K_\sigma * \mathbb{1}_B) dA_B \quad 6.2$$

↑
we convolved with just 1.

$$A_n^\sigma = \int d^2x \int d^2y \mathbb{1}_A(\vec{x}) K_\sigma(\vec{x} - \vec{x}') \mathbb{1}_B(\vec{x}') \quad 6.3$$

We need to be careful what K_σ we choose. If we choose Gaussian, it'll be a nightmare. We know this is not realistic anyway, so we can just use the Plummer model.

The 2D surface density from Wikipedia is

$$K_\sigma = \frac{\sigma^2}{\pi(\sigma^2 + |\vec{r}|^2)^2} \quad 6.4$$

but of course it doesn't give the potential or anything.

The vector field (ordinarily F) is given by

$$\vec{\nabla} \cdot \vec{E}_\sigma = K_\sigma \quad 6.5 \quad \text{and}$$

$$\vec{\nabla} \Phi_\sigma = \vec{E}_\sigma. \quad 6.6$$

$$\text{Let } \boxed{\Psi_B^\sigma(\vec{x}) \equiv (K_\sigma * \mathbb{1}_B)(\vec{x})} \quad 6.7$$

$$\begin{aligned} \Psi_B^\sigma(\vec{x}) &= \int d^2 y \, K_\sigma(\vec{y} - \vec{x}) \mathbb{1}_B(\vec{x}) \\ &= \int d^2 y \, (\vec{\nabla} \cdot \vec{E}_\sigma)(\vec{y} - \vec{x}) \mathbb{1}_B(\vec{x}) \end{aligned}$$

So from Gauss's theorem,

$$\boxed{\Psi_B^\sigma(\vec{x}) = \oint_{\partial B} \vec{E}_\sigma(\vec{y} - \vec{x}) \cdot \hat{n}_B(\vec{y}) d\ell_y} \quad 6.8$$

Since $\vec{E}_\sigma = \vec{\nabla} \Phi_\sigma$, we have

$$\boxed{\Psi_B^\sigma(\vec{x}) = \oint_{\partial B} (\vec{\nabla}_{\vec{x}} \Phi_\sigma)(\vec{y} - \vec{x}) \cdot \hat{n}_B(\vec{y}) d\ell_y} \quad 6.9$$

cite

And again, from Wikipedia, ^{Gradient theorem is}
 $\int_{\Omega} \vec{\nabla} u dV = \int_{\partial\Omega} u \vec{\nu} dS$ where
 $\vec{\nu}$ is the outward pointing unit normal vector to $\partial\Omega$.

Now,

$$A_n^\sigma = \int d^2x \Psi_B^\sigma(\vec{x})$$

$$A_n^\sigma = \int d^2x \oint_{\partial B} (\vec{\nabla}_{\vec{x}} \Phi_\sigma)(\vec{y} - \vec{x}) \cdot \hat{n}_B(\vec{y}) d\ell_y$$

and using the gradient theorem,

$$A_n^\sigma = \oint_{\partial A} d\ell_x \int_{\partial B} d\ell_y [-\Phi_\sigma(|\vec{y} - \vec{x}|)] \hat{n}_A(\vec{x}) \cdot \hat{n}_B(\vec{y}) \quad 6.10$$

7 Getting $\Phi_\sigma(r)$

We noted $\vec{\nabla} \cdot \vec{E} = K_\sigma$ which means

$$\boxed{\vec{E}_\sigma(\vec{r}) = C \frac{\vec{r}}{(\sigma^2 + |\vec{r}|^2)}} \quad 7.1$$

Quick check on this

$$\begin{aligned} \frac{\partial}{\partial x} C \frac{x}{\sigma^2 + |\vec{r}|^2} &= -C \frac{2x^2}{(\sigma^2 + |\vec{r}|^2)^2} \\ &\quad + C \frac{(\sigma^2 + |\vec{r}|^2)}{(\sigma^2 + |\vec{r}|^2)^2} \\ &= C \frac{\sigma^2 + y^2 - x^2}{(\sigma^2 + |\vec{r}|^2)^2} \end{aligned}$$

By symmetry,

$$\begin{aligned} \vec{\nabla} \cdot \vec{E}_\sigma &= C \left[\frac{2\sigma^2 + \cancel{y^2} - \cancel{x^2} + \cancel{x^2} - \cancel{y^2}}{(\sigma^2 + |\vec{r}|^2)^2} \right] \\ &= 2C \frac{\sigma^2}{\pi(\sigma^2 + |\vec{r}|^2)^2} = 2\pi C K_\sigma \end{aligned}$$

Therefore

$$\boxed{\vec{E}_\sigma(\vec{r}) = \frac{\vec{r}}{2\pi(r^2 + \sigma^2)} \quad 7.2}$$

The next consideration is that

$$\vec{\nabla} \Phi_\sigma = \vec{E}_\sigma(\vec{r}) \text{ so}$$

$$\Phi_\sigma = C \ln(r^2 + \sigma^2)$$

$$\left(C \frac{2x}{r^2 + \sigma^2}, C \frac{2y}{r^2 + \sigma^2} \right) \overset{\text{gives}}{=} \vec{E}_\sigma(\vec{r})$$

$$(2\pi)(2C) \vec{E}_\sigma(\vec{r}) = E_\sigma(\vec{r})$$

$$C = 1/4\pi \text{ and}$$

$$\boxed{\Phi_\sigma(r) = \frac{1}{4\pi} \ln(r^2 + \sigma^2)} \quad 7.3$$

Combining 6.10 and 7.3 gives

$$A_n^\sigma = -\frac{1}{4\pi} \oint_{2A} dl_x \oint_{2B} dl_y \ln(|\vec{x}-\vec{y}|^2 + \sigma^2) \hat{n}_A \cdot \hat{n}_B$$

7.4

We can apply this to our polygons

to get

$$A_n^\sigma = -\frac{1}{4\pi} \sum_{\alpha \in 2A} \sum_{\beta \in 2B} (\hat{n}_\alpha \cdot \hat{n}_\beta) \int_{e_\alpha} dl_x \int_{e_\beta} dl_y \ln(|\vec{x}-\vec{y}|^2 + \sigma^2)$$

7.5

Now what is

$$\int_{e_\alpha} dl_x \ln(|\vec{x}-\vec{y}|^2 + \sigma^2) ? \quad dl_x = |\vec{e}_\alpha| dt$$

Let's parameterize

$$\vec{X} = \vec{V}_\alpha + t \vec{e}_\alpha \text{ so that}$$

$$|\vec{e}_\alpha| \int_0^1 dt \ln(|\vec{V}_\alpha - \vec{y}|^2 + 2(\vec{V}_\alpha - \vec{y}) \cdot \vec{e}_\alpha t + |\vec{e}_\alpha|^2 t^2 + \sigma^2)$$

Let

$$\begin{aligned} a &= |\vec{e}_\alpha|^2 \\ b &= 2(\vec{V}_\alpha - \vec{y}) \cdot \vec{e}_\alpha \\ c &= |\vec{V}_\alpha - \vec{y}|^2 + \sigma^2 \end{aligned}$$

def 7.6

so

$$\mathcal{J} \equiv \int_0^1 dt \ln(at^2 + bt + c) \quad 7.65$$

We can define $u = \ln(at^2 + bt + c)$ and

$du = dt$ so

$$= t \ln(at^2 + bt + c) \Big|_0^1 - \int_0^1 dt \frac{2at^2 + bt}{at^2 + bt + c}$$

$$\mathcal{J} = \ln(a + b + c) - \int_0^1 dt \left[2 - \frac{bt + 2c}{at^2 + bt + c} \right]$$

We want to isolate the $2at + b$ term.

$$J = \ln(a+bt+c) - \int_0^1 dt \left[2 - \frac{bt+2c}{at^2+bt+c} \right]$$

Now we get rid of the bt by
 noting that $\frac{b}{2a} \frac{2at + (\frac{4ca}{b} - b) + b}{at^2+bt+c}$

$$J = \ln(a+bt+c) - \left[2 - \frac{b}{2a} \ln(at^2+bt+c) \right]_0^1 - \left(2c - \frac{b^2}{2a} \right) \int_0^1 \frac{dt}{at^2+bt+c}$$

$$J = \ln(a+bt+c) - 2 + \frac{b}{2a} \ln(a+bt+c) - \frac{b}{2a} \ln c + \left(2c - \frac{b^2}{2a} \right) \int_0^1 \frac{dt}{at^2+bt+c}$$

Let $\boxed{D \equiv \sqrt{4ac - b^2}}$ 7.7

We let $\boxed{\vec{w} \equiv \vec{v}_\alpha - \vec{y}}$ 7.8 so

$$4ac - b^2 = 4|\vec{e}_\alpha|^2 |\vec{w}|^2 - 4(\vec{w} \cdot \vec{e}_\alpha)^2$$

This means D is always real.

$$J = \left(1 + \frac{b}{2a}\right) \ln(a+b+c) - 2 - \frac{b}{2a} \ln c \\ + \frac{D}{a} \left[\arctan\left(\frac{2a+b}{D}\right) - \arctan\left(\frac{b}{D}\right) \right].$$

(note that the integral has a factor $\frac{2}{D}$. 7.9

$$A_n^\sigma = -\frac{1}{4\pi} \sum_{\alpha \in 2A} \sum_{\beta \in 2B} (\hat{n}_\alpha \cdot \hat{n}_\beta) |\vec{e}_\alpha| \underbrace{\int_{C_\beta} d\vec{y} J(\vec{y})}$$

We can integrate over $\vec{y}$ where $\underline{I}$

$$\vec{y}(s) = \vec{v}_\beta + \vec{e}_\beta s \text{ so}$$

$$\vec{w}(s) = \vec{v}_\alpha - \vec{v}_\beta - s\vec{e}_\beta$$

$$b(s) = 2\vec{w}(s) \cdot \vec{e}_\alpha$$

$$c(s) = |\vec{w}(s)|^2 + \sigma^2$$

$$b(s) = 2 \vec{\Delta} \cdot \vec{e}_\alpha - 2 \vec{e}_\alpha \cdot \vec{e}_\beta s$$

$$c(s) = |\vec{\Delta}|^2 - 2s \vec{\Delta} \cdot \vec{e}_\beta + s^2 |\vec{e}_\beta|^2 + \sigma^2 \quad 7.10$$

$$D(s) = \sqrt{4ac(s) - b(s)^2}$$

$$= 2 \sqrt{|\vec{e}_\alpha|^2 (|\vec{\omega}|^2 + \sigma^2) - |\vec{\omega} \cdot \vec{e}_\alpha|^2}$$

$$D(s) = 2 \sqrt{|\vec{e}_\alpha \times \vec{\omega}(s)|^2 + \sigma^2 |\vec{e}_\alpha|^2}$$

$$I = \int_{e_p} dl_g J(g)$$

$$D(s) = 2 \sqrt{|\vec{e}_\alpha \times \vec{\omega}(s)|^2 + \sigma^2} \quad 7.11$$

$$\frac{I}{|\vec{e}_\beta|} = \int_0^1 ds \left(1 + \frac{b(s)}{2a} \right) \ln(a + b(s) + c(s)) \quad 7.12$$

$$- 2 - \int_0^1 ds \frac{b(s)}{2a} \ln c(s)$$

$$+ \int_0^1 ds \frac{D(s)}{a} \left[\arctan\left(\frac{2a+b(s)}{D(s)}\right) - \arctan\left(\frac{b(s)}{D(s)}\right) \right]$$

Let's split this integral up further:

$$\frac{I}{|e_{\theta}|} = I_a - 2 - I_b + I_c - I_d$$

with

$$I_a = \int_0^1 ds \left(1 + \frac{b(s)}{2a} \right) \ln(a + b(s) + c(s))$$

$$I_b = \int_0^1 ds \frac{b(s)}{2a} \ln(c(s))$$

$$I_c = \int_0^1 ds \frac{D(s)}{a} \arctan\left(\frac{2a + b(s)}{D(s)}\right)$$

$$I_d = \int_0^1 ds \frac{D(s)}{a} \arctan\left(\frac{b(s)}{D(s)}\right)$$

7.13

Let's rotate our frame so that
 $\vec{e}_\alpha = \{|\vec{e}_\alpha|, 0\}$ and so

$$\begin{aligned} b &= 2|\vec{e}_\alpha|(\Delta^x - e_\beta^x s) \\ c &= |\vec{\Delta} - \vec{e}_\beta s|^2 + \sigma^2 \\ d &= 2|\vec{e}_\alpha| \sqrt{(\Delta^x - e_\beta^x s)^2 + \sigma^2} \end{aligned} \quad 7.14$$

$$a+b+c = |\vec{e}_\alpha|^2 + 2|\vec{e}_\alpha|(\Delta^x - e_\beta^x s) + (\Delta^x - e_\beta^x s)^2 + (\Delta^y - e_\beta^y s)^2 + \sigma^2$$

$$a+b+c = \left(|\vec{e}_\alpha| + \Delta^x - e_\beta^x s \right)^2 + (\Delta^y - e_\beta^y s)^2 + \sigma^2$$

Note that the above expression is nearly $C(s)$.
 If we replace $|\vec{e}_\alpha| + \Delta^x$ with Δ^x , we get C .

$$\text{Also } \frac{b}{2a} = \frac{\Delta^x - e_\beta^x s}{|\vec{e}_\alpha|} \text{ and } 1 + \frac{b}{2a} = \frac{\Delta^x - e_\beta^x s + |\vec{e}_\alpha|}{|\vec{e}_\alpha|}$$

So we see the generalization.

We can define a more general

structure :

$$J_1(p, s) \equiv \frac{p - e_p^x s}{|\vec{e}_\alpha|} \ln \left[\frac{(p - e_p^x s)^2 + (\Delta^y - e_p^y s)^2 + \sigma^2}{(\Delta^y - e_p^y s)^2 + \sigma^2} \right] \quad 7.15$$

$$\begin{aligned} \int_0^1 ds J_1(|\vec{e}_\alpha| + \Delta^x, s) &= I_a \\ \int_0^1 ds J_1(\Delta^x, s) &= I_b \end{aligned}$$

Similarly,

$$J_2(p, s) \equiv \frac{D(s)}{|\vec{e}_\alpha|^2} \arctan \left[\frac{2|\vec{e}_\alpha|(p - e_p^x s)}{D(s)} \right] \quad 7.16$$

so that

$$\begin{aligned} \int_0^1 ds J_2(|\vec{e}_\alpha| + \Delta^x, s) &= I_c \\ \int_0^1 ds J_2(\Delta^x, s) &= I_d \end{aligned}$$

Let's change variables. Define

$$\boxed{X \equiv \frac{1}{\sigma}(\Delta^y - e_{\beta^y} s)} \quad 7.17$$

$$\text{then } s = \frac{1}{e_{\beta^y}}(\Delta^y - \sigma X) \text{ and}$$

$$ds = -\frac{\sigma}{e_{\beta^y}} dx$$

and

$$P - e_{\beta^x} s = P - \frac{e_{\beta^x}}{e_{\beta^y}} \Delta^y + \frac{e_{\beta^x}}{e_{\beta^y}} \sigma X$$

$$\text{We can also let } \boxed{m \equiv \frac{e_{\beta^x}}{e_{\beta^y}}} \text{, the slope} \quad 7.18$$

$$\text{and } \boxed{V \equiv \frac{1}{\sigma}(P - m \Delta^y)} \text{, then y-intercept.} \quad 7.19$$

so

$$P - e_{\beta^x} s = \sigma(V + mX) \text{ and}$$

$$J_1(\nu, x) = \frac{\sigma}{|\vec{e}_\alpha|} (\nu + mx) \ln [(\nu + mx)^2 \sigma^2 + \sigma^2 x^2 + \sigma^2]$$

$$J_1(\nu, x) = \frac{\sigma}{|\vec{e}_\alpha|} (\nu + mx) \left\{ \ln [\nu^2 + 2m\nu x + m^2 x^2 + x^2 + 1] + 2 \ln \sigma \right\}$$

$$J_1(\nu, x) = \frac{2\sigma(\nu + mx)}{|\vec{e}_\alpha|} \ln \sigma + \frac{\sigma}{|\vec{e}_\alpha|} (\nu + mx) \ln [(1+m^2)x^2 + 2m\nu x + (1+\nu^2)]$$

7.20

$$X_0 \equiv \frac{\Delta^y}{\sigma} \quad \text{and} \quad X_1 \equiv \frac{1}{\sigma} (\Delta^y - e_\beta^y) \quad \text{so}$$

$$I_{\log} = \int_{X_0}^{X_1} \left(-\frac{\sigma}{e_\beta^y} \right) dx \left[\frac{2\sigma(\nu + mx)}{|\vec{e}_\alpha|} \ln \sigma - \int_{X_0}^{X_1} \frac{\sigma^2}{|\vec{e}_\alpha| e_\beta^y} (\nu + mx) \ln [(1+m^2)x^2 + 2m\nu x + (1+\nu^2)] \right]$$

7.21

Let's consider an integral of the form

$$\begin{array}{l|l} A = 1 + u^2 & \\ \hline B = 2uv & \\ \hline C = 1 + v^2 & \end{array}$$

$$\begin{aligned} 4AC - B^2 &= 4(1 + u^2)(1 + v^2) - 4u^2v^2 \\ &= 4[1 + v^2 + u^2 + \cancel{u^2v^2} - \cancel{u^2v^2}] \end{aligned}$$

$$4AC - B^2 = 4(1 + v^2 + u^2)$$

$$\begin{aligned} I_{\log} &= \frac{\sigma^2}{|\vec{e}_a| e_p^9} \left\{ \left[v(x_1 - x_2) + \frac{u}{2}(x_1^2 - x_2^2) \right] \ln \sigma^2 \right. \\ &\quad \left. + \int_{x_0}^{x_1} (v + ux) \ln(Ax^2 + Bx + C) \right\} \quad 7.22 \end{aligned}$$

Mathematica can handle the integral.

Now we do J_2 :

Recall that

$$D_0 = 2|\vec{e}_\alpha| \sqrt{(\Delta^y - e_\beta^y s)^2 + \sigma^2}$$

and

$$X = \frac{1}{\sigma}(\Delta^y - e_\beta^y s) \text{ so}$$

$$D(x) = 2|\vec{e}_\alpha| \sigma \sqrt{x^2 + 1} \quad 7.23$$

and

$$P - e_\beta^x s = (v + \ln x) \sigma \text{ so}$$

$$J_2(P, x) = \frac{2\sigma}{|\vec{e}_\alpha|} \sqrt{x^2 + 1} \arctan\left(\frac{v + \ln x}{\sqrt{x^2 + 1}}\right)$$

$$I_{\text{atan}}(P) \equiv - \frac{2\sigma^2}{|\vec{e}_\alpha| e_\beta^y} \int_{x_0}^{x_1} dx \sqrt{x^2 + 1} \arctan\left(\frac{v + \ln x}{\sqrt{x^2 + 1}}\right) \quad 7.24$$

We again do IBP so

$$u = \arctan\left(\frac{v+mx}{\sqrt{x^2+1}}\right) \Rightarrow du = dx \left[\frac{m-vx}{(Ax^2+Bx+C)\sqrt{x^2+1}} \right]$$

$$dv = \sqrt{x^2+1} dx$$

$$J_{atm}(\rho) = -\frac{2\sigma^2}{|\vec{e}_\alpha| e_\beta^y} \left\{ \arctan\left(\frac{v+mx}{\sqrt{x^2+1}}\right) \int \sqrt{x^2+1} dx \Big|_{x_0}^{x_1} - \int_{x_0}^{x_1} \left[\underbrace{\int \sqrt{x^2+1} dx}_{\frac{1}{2}(x\sqrt{x^2+1} + \operatorname{arcsinh}(x))} \left[\frac{m-vx}{\sqrt{x^2+1}(Ax^2+Bx+C)} \right] dx \right\} \quad 7.25$$

$$J_{atm}(\rho) = -\frac{2\sigma^2}{|\vec{e}_\alpha| e_\beta^y} \left\{ \frac{1}{2} \arctan\left(\frac{v+mx}{\sqrt{x^2+1}}\right) (x\sqrt{x^2+1} + \operatorname{arcsinh}(x)) \right. \\ \left. + \frac{1}{2} \int_{x_0}^{x_1} \left[\frac{vx^2-mx}{Ax^2+Bx+C} \right] + \frac{1}{2} \int_{x_0}^{x_1} dx \left(\frac{\operatorname{arcsinh}(x)}{\sqrt{x^2+1}} \right) x \right. \\ \left. \left(\frac{vx-m}{Ax^2+Bx+C} \right) \right\} \quad 7.26$$

The last piece is hard

$$J_{\text{arcsinh}} \equiv \frac{1}{2} \int_{x_0}^{x_1} dx \left(\frac{\text{arcsinh}(x)}{\sqrt{x^2+1}} \right) \times \left(\frac{Ux-m}{Ax^2+Bx+C} \right)$$

Let's separate this into two terms

so that

$$J_{\text{arcsinh}} = \frac{1}{2} \int_{x_0}^{x_1} dx \frac{\text{arcsinh}(x)}{\sqrt{x^2+1}} \left(\frac{\alpha_1}{x-\beta_1} + \frac{\alpha_2}{x-\beta_2} \right)$$

where

$$U = A(\alpha_1 + \alpha_2)$$

$$m = A(\alpha_1 \beta_2 + \alpha_2 \beta_1)$$

$$B = -(\beta_1 + \beta_2)A$$

$$C = \beta_1 \beta_2 A$$

We put this into Mathematica and get 2 symmetric solutions:

$$\alpha_1 = \frac{1}{2} \left[\frac{U - im \sqrt{U^2 + m^2 + 1}}{1 + m^2} \right]$$

$$\alpha_2 = \frac{1}{2} \left[\frac{U + im \sqrt{U^2 + m^2 + 1}}{1 + m^2} \right]$$

$$\beta_1 = - \left[\frac{mU + i \sqrt{U^2 + m^2 + 1}}{1 + m^2} \right]$$

$$\beta_2 = - \left[\frac{mU - i \sqrt{U^2 + m^2 + 1}}{1 + m^2} \right]$$

Depreciated

We then use partial fraction decomposition
on $R(y) \equiv \left(\frac{\alpha_1}{y-\beta_1} + \frac{\alpha_2}{y-\beta_2} + \frac{\alpha_3}{y-\beta_3} + \frac{\alpha_4}{y-\beta_4} \right)$ 7.30

We need to find α and β , but

$$I_{\text{arcsinh}} = \sum_{k=1}^4 \alpha_k \int_{y_0}^{y_1} \frac{\ln y}{y-\beta_k} dy \quad 7.31$$

If we let

$$L_k \equiv \int \frac{\ln y}{y-\beta_k} dy = \text{Li}_2\left(\frac{y}{\beta_k}\right) + \ln y \ln\left(1 - \frac{y}{\beta_k}\right) + C, \quad 7.32$$

$$\boxed{L_k = \text{Li}_2\left(\frac{y}{\beta_k}\right) + \ln y \ln\left(1 - \frac{y}{\beta_k}\right) \Big|_{y_0}^{y_1}} \quad 7.33$$

We have all the pieces except the partial fractions. We can try Mathematica's "Apart[R(y), y]"

We can put it all together from there in Mathematica. I will proceed with making a .nb that follows this notebook.

$$y^* = e^{t^*} = e^{\text{arcsinh}(x^*)} = e^{\text{arcsinh}\left(\frac{1}{\sigma}(\Delta^y - e_{\beta}^y)\right)}$$

$$\boxed{\begin{aligned} y_0 &= e^{\text{arcsinh}(\Delta^y/\sigma)} \\ y_1 &= e^{\text{arcsinh}\left(\frac{\Delta^y - e_{\beta}^y}{\sigma}\right)} \end{aligned}} \quad 7.34$$

8 Mollified Gradient

This derivation repeats the work from section 4 but with the new kernel.

We start with equation 6.10:

$$A_n^\sigma = - \oint_{\partial A} dx \oint_{\partial B} dy \Phi_\sigma(|\vec{y} - \vec{x}|) (\hat{n}_a(\vec{x}) \cdot \hat{n}_b(\vec{y}))$$

Next we note that the line element is

$$\boxed{\hat{N}_A^\alpha dx = \epsilon_{\alpha\beta} dx^\beta} \quad 8.1$$

and

$$\begin{aligned} N_A^\alpha dx N_B^\alpha dy &= \epsilon_{\alpha\beta} dx^\beta \epsilon_{\alpha\gamma} dy^\gamma \\ &= \delta_{\beta\gamma} dx^\beta dy^\gamma \\ &= dx^\gamma dy^\gamma. \end{aligned}$$

$$A_n^\sigma = - \oint_{2A} dx^\alpha \oint_{2B} dy^\alpha \Phi_\sigma(|\vec{y} - \vec{x}|) \quad 8.2$$

This is quite different from the situation in Section 4, but we apply the same technique.

Let $\vec{x} \mapsto \vec{x} + \delta\vec{x}$ so that

$$\delta A_n^\sigma = - \oint_{2B} dy^\alpha \oint_{2A} (\delta \Phi_\sigma(|\vec{y} - \vec{x}|) dx^\alpha + \Phi_\sigma d(\delta x^\alpha))$$

We know that $\delta \Phi_\sigma = \frac{\partial \Phi_\sigma}{\partial x^\beta} \delta x^\beta$

Therefore

$$\delta A_n^\sigma = - \oint_{2B} dy^\alpha \oint_{2A} \left[\frac{\partial \Phi_\sigma}{\partial x^\beta} \delta x^\beta dx^\alpha + \Phi_\sigma d(\delta x^\alpha) \right] \quad 8.3$$

As before, we integrate the second term by parts:

$$\int_{2A} d\delta x^\alpha \Phi_\sigma = \Phi_\sigma \delta x^\alpha \Big|_a^b - \int_{2A} \delta x^\alpha \frac{\partial \Phi_\sigma}{\partial x^\beta} dx^\beta$$

$$\begin{aligned} \delta A_n^\sigma &= - \int_{2B} dy^\alpha \int_{2A} \left[\frac{\partial \Phi_\sigma}{\partial x^\beta} \delta x^\beta dx^\alpha - \frac{\partial \Phi_\sigma}{\partial x^\gamma} \delta x^\alpha dx^\gamma \right] \\ &= - \int_{2B} \int_{2A} \left[\delta x^\beta dx^\alpha \frac{\partial \Phi_\sigma}{\partial x^\beta} dy^\alpha - \delta x^\alpha dx^\gamma \frac{\partial \Phi_\sigma}{\partial x^\gamma} dy^\alpha \right] \end{aligned}$$

Note that these are all dummy indices.

In the second term, consider $\alpha \rightarrow \beta$ and $\gamma \rightarrow \alpha$ so

$$\delta A_n^\sigma = - \int_{2A} dx^\alpha \int_{2B} \left[dy^\alpha \frac{\partial \Phi_\sigma}{\partial x^\beta} - dy^\beta \frac{\partial \Phi_\sigma}{\partial x^\alpha} \right] \delta x^\beta$$

8.4

Let's use equation 8.1 to put the dy back in:

$$\epsilon_{\alpha\beta} dy^\beta = \hat{n}_B^\alpha(\vec{y}) dy$$

Multiply both sides by $\epsilon_{\gamma\alpha}$ so

$$\epsilon_{\gamma\alpha} \epsilon_{\alpha\beta} dy^\beta = \epsilon_{\gamma\alpha} \hat{n}_B^\alpha dy$$

$$-dy^\gamma = -\epsilon_{\alpha\gamma} \hat{n}_B^\alpha dy$$

$$\delta A_n^\sigma = -\oint_{\partial A} dx^\alpha \oint_{\partial B} \left[\epsilon_{\gamma\alpha} \hat{n}_B^\gamma dy \frac{\partial \Phi_\sigma}{\partial x^\beta} - \epsilon_{\gamma\beta} \hat{n}_B^\gamma dy \frac{\partial \Phi_\sigma}{\partial x^\alpha} \right] \delta x^\beta$$

The term in brackets is antisymmetric

wrt α and β so consider $\alpha=0, \beta=1$

$$\epsilon_{10} \hat{n}_B^1 dy \frac{\partial \Phi_\sigma}{\partial x^1} - \epsilon_{01} \hat{n}_B^0 dy \frac{\partial \Phi_\sigma}{\partial x^0} \text{ and}$$

since $\epsilon_{10} = -1$ and this is antisymmetric for $\alpha=1, \beta=0$,

$$\boxed{\delta A_n^\sigma = -\oint_{\partial A} dx^\alpha \oint_{\partial B} dy \epsilon_{\alpha\beta} \hat{n}_B^\gamma \frac{\partial \Phi_\sigma}{\partial x^\gamma} \delta x^\beta} \quad 8.5$$

From equation 6.9, we have

$$\delta A_n^\sigma = - \oint_{2A} dx^\alpha \delta x^\beta \epsilon_{\alpha\beta} \Phi_B(\vec{x}) \quad 8.6$$

We can traverse the edges on this outer integral over $2A$.

If we parameterize as before:

$$\begin{aligned} \vec{X}_k(s) &= \vec{V}_k + s \vec{E}_k \\ &= (1-s)\vec{V}_k + s\vec{V}_{Z(k)} \end{aligned}$$

and $dx_k^\alpha = e_k^\alpha ds$ so

$$\delta A_n^\sigma = - \sum_{k \in 2A} \sum_{\beta} \int_0^1 e_k^\alpha ds [(1-s)\delta V_k^\beta + s\delta V_{Z(k)}^\beta] \epsilon_{\alpha\beta} \Phi_B(\vec{X}_k(s))$$

If we let

$$W_k^0 \equiv \int_0^1 ds (1-s) \Phi_B(\vec{X}_k(s)) \quad 8.7$$

$$W_k^1 \equiv \int_0^1 ds s \Phi_B(\vec{X}_k(s)) \quad 8.8$$

then

$$\delta A_n^\sigma = - \sum_{k \in 2A} \sum_{\beta} \left[W_{1k}^0 \delta_{km} + W_{1k}^1 \delta_{z(k)m} \right] \epsilon_{\alpha\beta} e_k^\alpha \delta V_m^\beta$$

meaning

$$\frac{\delta A_n^\sigma}{\delta V_m^\beta} = - \sum_{k \in 2A} \left[W_{1k}^0 \delta_{km} + W_{1k}^1 \delta_{z(k)m} \right] \epsilon_{\alpha\beta} e_k^\alpha$$

$$\boxed{\frac{\delta A_n^\sigma}{\delta V_m^\beta} = \epsilon_{\beta\alpha} \left[W_m^0 e_m^\alpha + W_{z'(m)}^1 e_{z'(m)}^\alpha \right]}$$

8.9

Recall:

$$\Psi_B^\sigma(\vec{x}) = \oint_{\partial B} \vec{E}(\vec{y} - \vec{x}) \cdot \hat{n}_B(\vec{y}) d\ell_y$$

and again use $\hat{n}_B^\alpha d\ell_y = \epsilon_{\alpha\beta} dy^\beta$

$$\Psi_B^\sigma(\vec{x}) = \oint_{\partial B} \vec{E}_\sigma(\vec{y} - \vec{x}) \epsilon_{\alpha\beta} dy^\beta$$

Now we parameterize for an edge

$$\vec{y}_k(s) - \vec{x} = \vec{w}_k + \vec{e}_k s - \vec{x}$$

$$\vec{y}_k(s) - \vec{x} = -\vec{w}_k + \vec{e}_k s$$

$$\text{then } d\vec{y}_k = \vec{e}_k ds.$$

Using the definition for E_σ ,

$$\Psi_k^\sigma(\vec{x}) = \int_0^1 ds \frac{-\vec{w}_k^\alpha + \vec{e}_k^\alpha s}{2\pi(|\vec{w}_k - \vec{e}_k s|^2 + \sigma^2)} \epsilon_{\alpha\beta} e_k^\beta \quad 8.10$$

Well, doesn't 8.10 look familiar?

$$(\vec{w}_k - \vec{e}_k s)^2 = |\vec{w}_k|^2 - 2 \vec{w}_k \cdot \vec{e}_k s + |\vec{e}_k|^2 s^2$$

and like before,

$$a_k \equiv |\vec{e}_k|^2; \quad b_k \equiv -2 \vec{w}_k \cdot \vec{e}_k; \quad c_k \equiv |\vec{w}_k|^2 + \sigma^2$$

gives

$$\Psi_k^\sigma(\vec{x}) = - \int_0^1 ds \frac{w_k^\alpha \epsilon_{\alpha\beta} e_k^\beta}{2\pi(a_k s^2 + b_k s + c_k)}$$

as before,

$$\Psi_k^\sigma(\vec{x}) \equiv - \frac{w_k^\alpha \epsilon_{\alpha\beta} e_k^\beta}{\pi D_k} \left[\arctan\left(\frac{2a_k + b_k}{D_k}\right) - \arctan\left(\frac{b_k}{D_k}\right) \right]$$

8.11

$$\Psi_B^\sigma(\vec{x}) = \sum_{k \in B} \Psi_k^\sigma(\vec{x})$$

8.12

$$W_k^0 \equiv \int_0^1 ds (1-s) \Phi_B(\vec{x}_k(s))$$

$$| \quad W_k^1 \equiv \int_0^1 ds s \Phi_B(\vec{x}_k(s))$$